

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 23/01/2026**
Time: 3 Hrs **GTM - 23** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Calculate the energy stored in an imaginary cubical volume of side a placed in front of an infinitely large nonconducting sheet of uniform charge density σ.

A) $\frac{\sigma^2 a^3}{8\epsilon_0}$ B) $\frac{\sigma^2 a^2}{4\epsilon_0}$ C) $\frac{\sigma^2 a^3}{4\epsilon_0}$ D) $\frac{\sigma^2 a^2}{8\epsilon_0}$

2. A convergent lens of power 6 D is combined with a diverging lens of -2 D. Find the power and focal length of the combination.

A) $15cm$ B) $25cm$ C) $50cm$ D) $125cm$

3. Calculate the pressure inside a small air bubble of 0.1 mm radius situated just below the water surface. Assume the surface tension of water to be $7.2\times10^{-2}\,Nm^{-1}$ and atmospheric pressure to be $1.013\times10^{5}\,Nm^{-2}$

A) $1.027\times10^{5}\,Nm^{-2}$ B) $1.027\times10^{6}\,Nm^{-2}$

C) $1.041\times10^{5}\,Nm^{-2}$ D) $1.041\times10^{6}\,Nm^{-2}$

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4. The diameter and height of a cylinder are measured by a meter scale to be $12.6 \pm 0.1 cm$ *and* $34.2 \pm 0.1 cm$, respectively. What will be the value of its volume in appropriate significant figures?

A) $4264 \pm 81 cm^3$ B) $4300 \pm 80 cm^3$

C) $4260 \pm 80 cm^3$ D) $4264.4 \pm 81.0 cm^3$

5. Calculate the energy of a H-atom in the first excited state, if the potential energy is assumed to be zero in the ground state.

A) $-27.2eV$ B) $23.8eV$ C) $-23.8eV$ D) $27.2eV$

6. Consider the following statements:

a) The junction area of solar cell is made very narrow compared to a photo diode.

b) Solar cells are not connected with any external bias.

c) LED is made of lightly doped p-n junction.

d) Increase of forward current results in continuous increase of LED light intensity.

e) LEDs have to be connected in forward bias for emission of light.

A) b, d, e Only B) a, c Only C) a, c, e Only D) b, e Only

7. How much work will be done in increasing the diameter of a soap bubble from 2cm to 5cm. Surface tension of soap solution is $3 \times 10^{-2} Nm^{-1}$

A) 3.96×10^{-4} J B) 3.96×10^{-10} J C) 3.96×10^{-8} J D) 3.96×10^{-6} J

8. A particle is projected at time t=0 from a point 'O' with a speed 'u' at an angle 45^0 to horizontal. Find the angular momentum of the particle at time t=u/g.

A) $\frac{-mu^3}{2\sqrt{2}g}\hat{k}$ B) $\frac{-mu^3}{\sqrt{2}g}\hat{k}$ C) $\frac{-mu^2}{2\sqrt{2}g}\hat{k}$ D) $\frac{-mu^3}{4\sqrt{2}g}\hat{k}$

9. Light from a source consist of two wavelengths $\lambda_1 = 6500 \overset{0}{\text{A}}$ *and* $\lambda_2 = 5200 \overset{0}{\text{A}}$ If the separation between the sources from each other is 6.5 mm and that from the screen is 2 m, find the minimum value of $y(\neq 0)$ where the maxima of both the wavelengths coincide.

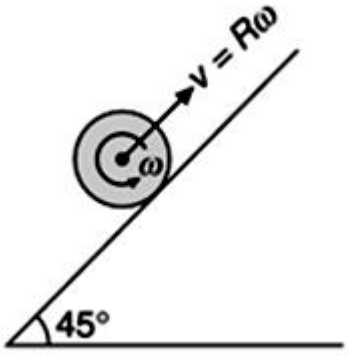
A) 0.2 mm B) 0.4 mm C) 0.8 mm D) 1.6 mm

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10. A vehicle whose wheel track is 1.6m wide and whose centre of gravity is 1 m above the road, centred between the wheels, takes a curve whose radius is 50 m, on a level road. Taking $g=10ms^{-2}$, find the speed at which the inner wheel would leave the road.

A) $5m/s$ B) $10m/s$ C) $15m/s$ D) $20m/s$

11. A solid sphere of radius R is projected along an inclined plane as shown in Figure.



If coefficient of friction is $\mu = 0.5$ then calculate the time when linear velocity of the sphere becomes zero.

A) $\frac{2\sqrt{2}R\omega}{5g}$ B) $\frac{4\sqrt{2}R\omega}{5g}$ C) $\frac{5g}{4\sqrt{2}}$ D) $\frac{4\sqrt{2}}{5g}$

12. A glass convex lens of refractive index (3/2) has a focal length equal to 0.3m, Find the focal length of the lens if its is immersed in water of refractive index (4/3)?

A) 0.6 m B) 1.2 m C) 2.4 m D) 4.8 m

13. Two particles move parallel to the x-axis about the origin with the same frequency and amplitude a. At a certain instant they are found at distances $\frac{a}{3}$ from the origin on opposite sides but their velocity is found to be in the same direction. Calculate the phase different between the two particles.

A) $\cos^{-1}\left(\frac{7}{9}\right)$ B) $\cos^{-1}\left(\frac{5}{7}\right)$ C) $\cos^{-1}\left(\frac{1}{3}\right)$ D) $\cos^{-1}\left(\frac{5}{9}\right)$

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14. A satellite of mass m is moving in a circular orbit of radius r. Due to atmospheric drag if it loses energy at a constant rate W. Find the time in which satellite will fall to the surface of earth. The mass of the earth is M and radius R.

A) $\frac{GMm}{2W}\left(\frac{1}{R}-\frac{1}{r}\right)$ B) $\frac{GMm}{2WR}\left(\frac{1}{R}\frac{-1}{r}\right)$

C) $\frac{GMm}{WR}\left(\frac{1}{R}\frac{-1}{r}\right)$ D) $\frac{GMm}{W}\left(\frac{1}{R}\frac{-1}{r}\right)$

15. Two thin symmetrical lenses of different nature and of different material have equal radii of curvature R=15cm. The lenses are put close together and immersed in water $\left(\mu_w=\frac{4}{3}\right)$ The focal length of the system in water is 30cm. The difference between refractive indices of two lenses is

A) $\frac{1}{3}$ B) $\frac{2}{3}$ C) $\frac{4}{3}$ D) $\frac{8}{3}$

16. A current is made of two components a dc component $i_1=3A$ and an ac component $i_2=4\sqrt{2}\sin\omega t$. Find the reading of hot wire ammeter?

A) 4 B) 5 C) 7 D) 1

17. Find the ratio of de Broglie wavelength of proton and α - particle which having accelerated through same potential difference.

A) $\sqrt{2}$ B) $\frac{1}{\sqrt{2}}$ C) $2\sqrt{2}$ D) $\frac{1}{2\sqrt{2}}$

18. Two conductors carrying equal and opposite charges produce a non-uniform electric field along X- axis given by $E=\frac{Q}{\varepsilon_0 A}(1+Bx^2)$ where A and B constants. Separation between the conductors along X-axis is X. Find the capacitance of the capacitor formed.

A) $\frac{\in_0 A}{x\left(1+\frac{Bx^2}{3}\right)}$ B) $\frac{\in_0 A}{x^2\left(1+\frac{Bx^3}{3}\right)}$

C) $\frac{3\in_0 A}{x\left(1+Bx^2\right)}$ D) $\frac{\in_0 A}{x^3\left(1+\frac{Bx^3}{3}\right)}$

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19. An object is displaced from a point A (0, 0, 0) to B (1m, 1m, 1m) under force $\vec{F} = (y\hat{i} + x\hat{j})N$. Calculate the work done by this force in this process.

A) 1J B) 4J C) 2J D) 5J

20. An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statements are true?

a) The mean free path of the molecules decreases.

b) The mean collision time between the molecules decreases.

c) The mean free path remains unchanged.

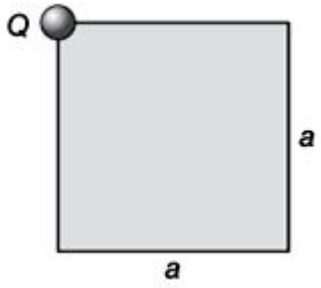
d) The mean collision time remains unchanged.

A) c and d B) a and b C) a and d D) b and c

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

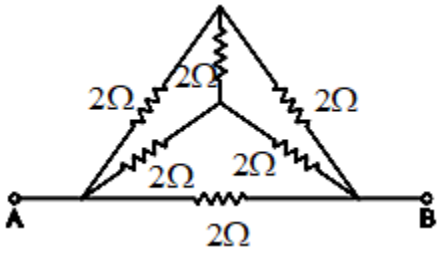
21. A point charge Q is placed at the corner of a square plate of side a, then find the flux through the square plate is___



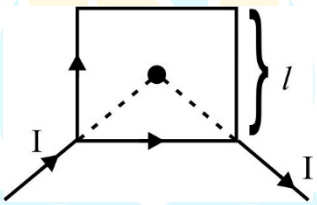
22. Length of a thin cylinder as measured by a Vernier caliper having least count 0.01 cm is 3.25 cm and its radius of cross-section is measured by a screw gauge having least count 0.01 mm as 2.75 mm. The percentage error in the measurement of volume of the cylinder will be____

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23. Six resistors of each 2 ohms are connected as shown in figure. The resultant resistance between A and B is.



24. The magnitude of magnetic field at centre of the square loop of side l as shown in figure is $x\left(\frac{\sqrt{2}\mu_0 I}{\pi l}\right)$. Find x



25. The temperature of 2 moles of an ideal monoatomic gas is increased by $40^0 C$ at constant pressure. The total heat supplied and the change in internal energy are Q_1 and Q_2, respectively. If $\frac{Q_1}{Q_2} = \frac{x}{6}$

Then the value of x is ____.

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The E.M.F of cell at $25^0\,C$ is

$$Zn/\underset{(0.01M)}{Zn^{+2}}//\underset{(0.1M)}{H^+}/\underset{(1bar)}{H_2}\quad E^0_{(Zn/Zn^{+2})}=0.77V$$

A) $-0.77V$ B) $0.77V$ C) $1.34V$ D) $2.14V$

27. The number of spectral lines produced in the visible region of H – spectrum, when electron jumps from 5th shell to 1st shell

A) 10 B) 3 C) 4 D) 2

28. The number of ions precipitated on treatment of $100\,ml$ of $0.1M$ solution of $CoCl_3.6H_2O$ with excess $AgNO_3$ are 1.2×10^{22}. The complex is

A) $\left[Co(H_2O)_6\right]Cl_3$ B) $\left[Co(H_2O)_5\,Cl\right]Cl_2H_2O$

C) $\left[Co(H_2O)_4\,Cl_2\right]Cl.2H_2O$ D) $\left[Co(H_2O)_3\,Cl_3\right]3H_2O$

29. Which of the following is most stable carbocation

A) $Ph_3\overset{\oplus}{C}$ B) $H_5C_6-\overset{\oplus}{C}H_2$

C) $H_5C_6-\overset{\oplus}{C}H_3-C_6H_5$ D) $CH_3-\overset{\oplus}{C}H(CH_3)-CH_3$

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30. Select the correct statements

I) The dipole moment order of the following molecules CH_4, CCl_4 & $CHCl_3$ is $CHCl_3 > CH_4 = CCl_4$

II) NH_3 has higher dipole moment than NF_3

III) The inter molecular forces involved in non-polar molecules are Dipole - Induced dipole forces

IV) Both CO_2 and SO_2 are non-polar molecules

A) II & IV B) III & IV C) I, II & III D) I & II

31. Magnitude of hydration energy is maximum for

A) Mn^{2+} B) Co^{2+} C) Cu^{2+} D) V^{2+}

32. Solubility product of $M(OH)_2$, if its saturated solution has $pH = 10$ at $25^0 C$

A) 5×10^{-12} B) 5×10^{-13} C) 5×10^{-30} D) 5×10^{-7}

33. For a particular reversible reaction at temperature T. ΔH and ΔS were found to be both $+ve$. If T_e is the temperature at equilibrium, the reaction would be spontaneous when

A) $T = T_e$ B) $T_e > T$ C) $T > T_e$ D) T_e is 5 times T

34. An organic compound contains 48% carbon, 8% hydrogen and remaining is of oxygen, molecular formula of an organic compound if its molecular weight 148 is_____

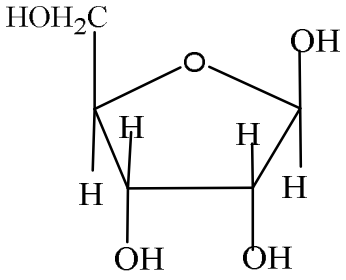
A) $C_6H_{12}O_4$ B) $C_3H_6O_2$ C) C_2H_6O D) $C_6H_{12}O_6$

35. Which of the following species do not show disproportionation reaction

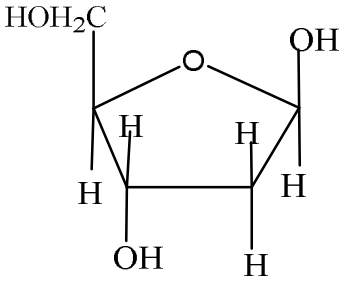
A) ClO_4^- B) ClO_3^- C) ClO_2^- D) ClO^-

36. DNA and RNA both contain pentose sugar, but composition of sugar may be different. Two pentose sugars A and B are given below

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(A)



(B)

Correct statement is

A) Sugar A is present in DNA while sugar B is present in RNA

B) Sugar A is present in RNA while sugar Bis present in DNA

C) Sugar B is present in both RNA and DNA

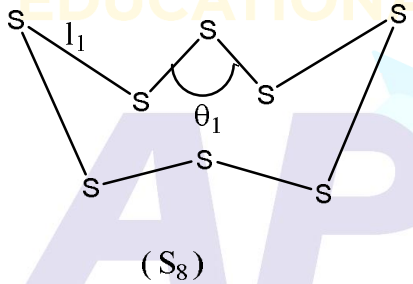
D) sugar A is present in both RNA and DNA

37. Which of the following conversion is incorrect in the presence of acidified $KMnO_4$ solution?

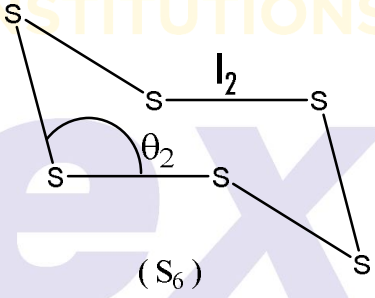
A) $I^- \rightarrow I_2$ B) $C_2O_4^{2-} \rightarrow CO_2$

C) $H_2SO_4 \rightarrow SO_2$ D) $H_2S \rightarrow S$

38. Consider the structures of S_8 *and* S_6 given below



(S_8)



(S_6)

Choose the correct combination about lengths l_1 and l_2 ; θ_1 *and* θ_2

A) $l_1 = l_2$ *and* $\theta_1 = \theta_2$ B) $l_1 < l_2$ *and* $\theta_1 > \theta_2$

C) $l_1 > l_2$ *and* $\theta_1 < \theta_2$ D) $l_1 < l_2$ *and* $\theta_1 < \theta_2$

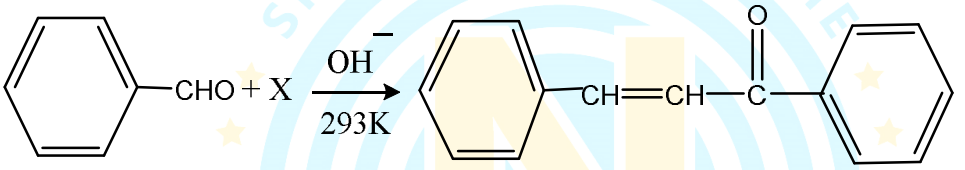
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39. If α is the degree of dissociation at equilibrium, then K_p of following will be (Take total pressure (P) at equilibrium as $0.25atm$)

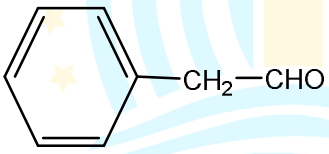
$N_2O_{4(g)} \rightleftarrows 2NO_{2(g)}$

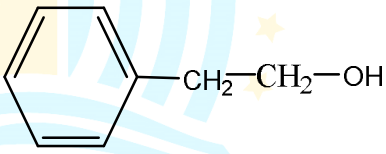
A) $\frac{\alpha^2}{1-\alpha^2}$ B) $\frac{4\alpha^2}{1-\alpha^2}$ C) $4\alpha^2$ D) $2\alpha^2$

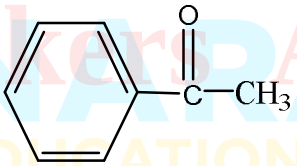
40.

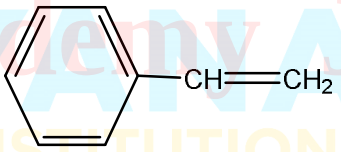


Identify 'X'

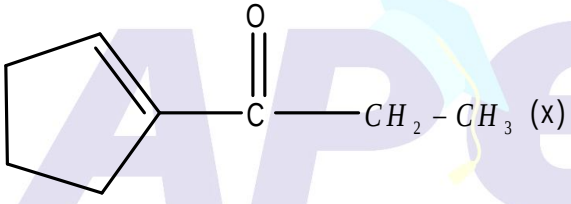
A) 

B) 

C) 

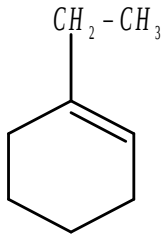
D) 

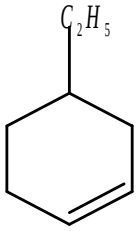
41. Arun has been asked to synthesize the molecule

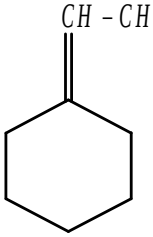


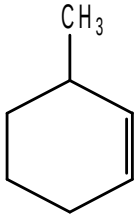
.He thought of preparing the molecule using an aldol condensation reaction. He found a few cyclic alkenes in his laboratory. He thought of preforming ozonolysis reaction on alkene to forma product known as Di carbonyl compound followed by Aldol reaction to prepare 'X'. Predict the suitable alkene that can lead to the formation of 'X'.

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A) 

B) 

C) 

D) 

42. Which of the following is incorrect statement

A) KJeldahl method is not applicable to compounds containing nitrogen in nitro and azo groups and nitrogen present in pyridine

B) Carius method is used for the estimation of halogens

C) Phosphorous present in the organic compound is oxidized to phosphoric acid and it is precipitated as ammonium phosphomolybdate by adding ammonia and ammonium molybdate

D) Dumas method is used for the estimation of halogens.

43. Match List-I with List -II

List-I	List-II
A) $\psi_{MO}=\psi_A-\psi_B$	I) Dipole moment
B) $\mu=Q\times r$	II) Bonding molecular orbital
C) $\frac{N_b-N_a}{2}$	III) Anti-bonding molecular orbital
D) $\psi_{MO}=\psi_A+\psi_B$	IV) Bond order

Choose the correct answer from options below.

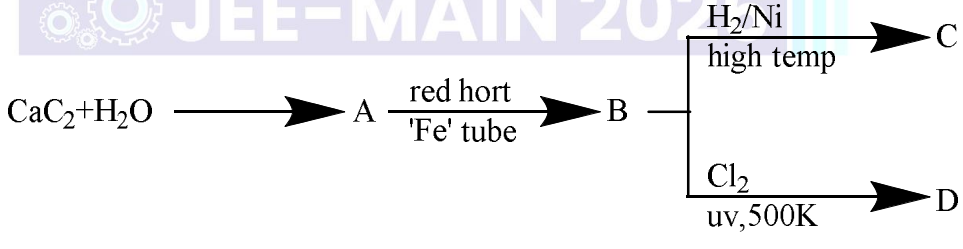
A) A-II, B-I, C-IV, D-III

B) A-III, B-IV, C-I, D-II

C) A-III, B-I, C-IV, D-II

D) A-III, B-IV, C-II, D-I

44.

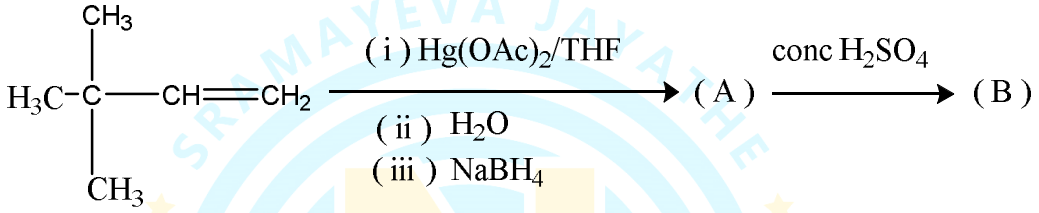


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The number of Sp^3 hybridized carbons in compound 'C' is x and that of in 'D' is y, therefore $x+y$ is____

A) 0 B) 6 C) 12 D) 18

45. How many α-hydrogens are present in major product (B)?



A) 0 B) 2 C) 6 D) 12

SECTION – II
(Numerical Value Answer Type)

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46. Following equilibrium exists in a 10 L container. Initially 100g of $CaCO_3$ was taken

$CaCO_{3(g)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ $\left(K_P = 2.16\ atm\ at\ 790^oC\right)$

The percentage dissociation of $CaCO_3$ at equilibrium is …… (round off nearest integer)

47. The enthalpy of combustion of propane, graphite and dihydrogen at 298 K are $-2220.0 KJ\ mol^{-1}, -393.5 KJ\ mol^{-1}$ and $-285.8 KJ\ mol^{-1}$ respectively .The magnitude of enthalpy of formation of propane $\left(C_3H_8\right)$ is……. $KJ\ mol^{-1}$ (nearest integer)

48. When $800\,ml\ of\ 0.5M$ nitric acid is heated in a beaker, its volume is reduced to half and $11.5\,g$ of nitric acid is evaporated. The molarity of the remining nitric acid solution is $X\times10^{-2}\,M$ (Molar mass of nitric acid is $63 g\ mol^{-1}$). Then x is_____ (nearest integer)

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49. Out of the following X is number of salts which have yellow or orange colour precipitates. Then value of X^2 is .....

CdS, ZnS, As_2S_3, As_2S_5, CoS, $BaCO_3$, PbS, SnS_2, AgI, PbI_2.

50. A metal oxide has the formula M_2O_3.This oxide with hydrogen to produce M and H_2O according to the reaction(unbalanced) $M_2O_3 + H_2 \rightarrow M + H_2O$.If $32g$ M_2O_3 required $1.2g$ of hydrogen for complete reduction. The atomic mass of metal M is $(in\ gm/mol)$ _____ (nearest integer)

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. The circle with equation $x^2 + y^2 = 1$ intersects the line $y = 7x + 5$ at two distinct points A and B. Let C be the point at which the positive x-axis intersects the circle. The $\angle ACB$ is

A) $\tan^{-1}\left(\frac{4}{3}\right)$ B) $\tan^{-1}\left(\frac{3}{4}\right)$ C) $\frac{\pi}{4}$ D) $\tan^{-1}\left(\frac{3}{2}\right)$

52. The point on the line $\frac{x-2}{1} = \frac{y+3}{-2} = \frac{z+5}{-2}$ at a distance of 6 from the point $(2,-3,-5)$ is $(0, y_1, z_1)$ then the value of $(y_1 + z_1)^2$ is

A) 4 B) 9 C) 0 D) 25

53. A chord is drawn passing through P (2,2) on the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ such that it intersects the ellipse at A and B. Then the maximum value of $PA \times PB$ is

A) $\frac{61}{4}$ B) $\frac{59}{4}$ C) $\frac{71}{4}$ D) $\frac{63}{4}$

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54. The pair of the lines whose direction cosines are given by the equations $3l+m+5n=0$, $6mn-2nl+5lm=0$, are

A) Parallel B) Perpendicular

C) Inclined at an angle $\cos^{-1}\left(\frac{1}{6}\right)$ D) Inclined at an angle $\cos^{-1}\left(\frac{1}{3}\right)$

55. A triangle ABC with vertices A (-1,0), B (-2,3/4) and C (-3, -7/6) has its orthocentre H. Then the orthocentre of triangle BCH will be

A) (-3, -2) B) (1,3) C) (-1,2) D) $(-1,0)$

56. Let 1, ω and ω^2 be the cube roots of unity. The least possible degree of a polynomial, with real coefficients, having $2\omega^2$, $3+4\omega$, $3+4\omega^2$ and $5-\omega-\omega^2$ as roots is _____

A) 3 B) 4 C) 5 D) 6

57. The mean of the data set comprising of 16 observations is 16. If one of the observation valued 16 is deleted and three new observations valued 3,4 and 5 are added to the data, then the mean of the resultant data, is:

A) 15.8 B) 14.0 C) 16.8 D) 16.0

58. Let a function $f(x)$ is defined in $[-2,2]$ as

$$f(x)=\begin{cases}\{x\}, & -2\le x<-1\\ |\operatorname{sgn} x| & -1\le x\le 1, \text{where } \{x\} \text{ and } \operatorname{sgn} x\\ \{-x\} & 1<x\le 2\end{cases}$$

denote the fractional part and signum functions, respectively. Then the area bounded by the graph of $f(x)$ and the x-axis is ____.

A) 3 B) 4 C) 5 D) 6

59. Let $f(x)=\lim\limits_{n\to\infty}\dfrac{2x^{2n}\sin\frac{1}{x}+x}{1+x^{2n}}$, then which of the following is not true

A) $\lim\limits_{x\to\infty} xf(x)=2$ B) $\lim\limits_{x\to 1} f(x)$ does not exist

C) $\lim\limits_{x\to 0} f(x)$ does not exist D) $\lim\limits_{x\to-\infty} f(x)$ is equal to zero

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60. Let $I_n = \int_0^1 x^n \sqrt{1-x^2}\ dx$, then $\lim_{n\to\infty} \frac{I_n}{I_{n-2}} =$

A) 2 B) 1 C) -1 D) -2

61. A biased coin with probability $P(0<P<1)$ of falling tail is tossed until a tail appears for the first time. If the probability that tails comes in odd number of trails is $2/3$, then P equals

A) $\frac{1}{4}$ B) $\frac{1}{3}$ C) $\frac{3}{4}$ D) $\frac{1}{2}$

62. Let f be a real valued function such that $f(x)+2f\left(\frac{2002}{x}\right)=3x$ for all $x>0$. The value of $f(2)$ is

A) 1000 B) 2000 C) 3000 D) 4000

63. The solution of differential equation $(1-xy+x^2y^2)dx = x^2dy$ is

A) $\tan xy = \log|cx|$ B) $\tan(y/x) = \tan\log|cx|$

C) $xy = \tan\log|cx|$ D) $\tan(x/y) = \tan\log|cx|$

64. Area of the region bounded by the curve $y=\frac{16-x^2}{4}$ and $y=\sec^{-1}\left[-\sin^2 x\right]$ (where $[\bullet]$ denotes the greatest integer function) is

A) $\frac{1}{3}(4-\pi)^{3/2}$ B) $8(4-\pi)^{3/2}$

C) $\frac{8}{3}(4-\pi)^{3/2}$ D) $\frac{8}{3}(4-\pi)^{1/2}$

65. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}|=1$, $|\vec{b}|=4, \vec{a}.\vec{b}=2$. If $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is

A) $\frac{\pi}{3}$ B) $\frac{\pi}{6}$ C) $\frac{3\pi}{4}$ D) $\frac{5\pi}{6}$

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66. In a certain town, 25% of the families own a phone and 15% own a car; 65% families own neither a phone nor a car and 2,000 families own both a car and a phone. Consider the following three statements:

a) 5% families own both a car and a phone

b) 35% families own either a car or a phone

c) 40, 000 families live in the town

Then,

A) Only (a) and (c) are correct. B) Only (b) and (c) are correct.

C) All (a), (b) and (c) are correct. D) Only (a) and (b) are correct.

67. $\frac{\alpha^3}{2}\cos ec^2\left(\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right)+\frac{\beta^3}{2}\sec^2\left(\frac{1}{2}\tan^{-1}\frac{\beta}{\alpha}\right)$ is equal to

A) $(\alpha-\beta)\left(\alpha^2+\beta^2\right)$ B) $(\alpha+\beta)\left(\alpha^2-\beta^2\right)$

C) $(\alpha+\beta)\left(\alpha^2+\beta^2\right)$ D) $\left(\alpha^3+\beta^3\right)(\alpha-\beta)$

68. A is a 2×2 matrix such that $A\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}-1\\2\end{bmatrix}$ and $A^2\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix}$. The sum of the elements of A is

A) – 1 B) 0 C) 2 D) 5

69. The order of the differential equation whose general solution is $y=c_1\cos 2x+c_2\cos^2 x+c_3\sin^2 x+c_4$ is

A) 2 B) 4 C) 3 D) 1

70. The number of positive 6-digit integers which is divisible by 9 and five of its digits are $1,0,0,0,5$ is

A) 60 B) 120 C) 180 D) 210

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. If $A=\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ and X is a non-zero column matrix such that $AX=\lambda X$, where λ is a scalar, then the sum of the values of λ is equal to

72. If $6^{83}+8^{83}$ is divided by 49, then the remainder is

73. If α,β are the roots of the quadratic equation $x^2-(3+2^{\sqrt{\log_2^3}}-3^{\sqrt{\log_3^2}})x-2(3^{\log_3^2}-2^{\log_2^3})=0$ then the value of $\alpha^2+\alpha\beta+\beta^2$ is equal to

74. If $a_1,a_2,a_3.....a_{4001}$ are terms of an AP such that $\frac{1}{a_1a_2}+\frac{1}{a_2a_3}+.....+\frac{1}{a_{4000}a_{4001}}=10$ and $a_2+a_{4000}=50$, then $|a_1-a_{4001}|$ is equal to

75. If $\lim\limits_{x\to 0}\frac{x^3}{\sqrt{(a+x)}(bx-\sin x)}=1$, $a>0$ Then $a+b$ is equal to

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